NOT MEASUREMENT SENSITIVE

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DEPARTMENT OF DEFENSE HANDBOOK

ELECTROSTATIC DISCHARGE PROTECTIVE PACKAGING

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AMSC N/A **AREA PACK**

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. Military packaging has experienced many changes in the packaging of electrical devices. Many of the new electronic parts and devices are highly susceptible to damage from static electricity, even at levels that can be neither seen nor felt. Electrostatic Discharge (ESD) affects many components such as transistors, resistors, integrated circuits, and many types of semiconductor devices. Static charge can degrade or destroy electronic devices during shipping, handling, and installation. Often it takes only a small charge to cause this damage. In fact, most of today's new semiconductors, with their denser geometries, have a static sensitivity of 100 volts or less. This means that almost any moving object can generate a dangerous charge in an unprotected environment. Note that an ESD event (with or without a resulting spark), may not result in immediate failure or evident damage to the Electrostatic Discharge Sensitive (ESDS) item. The ESDS item could be partially degraded whereby it may function normally for a period-of-time, but experiences unpredictable intermittent or permanent failures at a future time. This is known as a latent defect.

3. It is essential that everyone involved in the acquisition, handling, transporting, and storing of Electrostatic Discharge Sensitive (ESDS) items be concerned about ESD. All ESDS items should be packaged, shipped, and stored in ESD protective materials, including condition code F returns (economically repairable material, which requires repair, reconditioning, or overhaul).

4. Reducing damage to ESDS items begins with understanding the problem. This handbook is intended to provide guidelines on precautionary measures for reducing failures as they relate specifically to packaging and handling of ESDS items.

5. Comments, suggestions, or questions on this document should be addressed to Army Sustainment Command, Packaging, Storage, and Containerization Center, ATTN: AMAS-SPI-P, 11 Hap Arnold Blvd, Tobyhanna, PA 18466-5097 or emailed to usarmy.tyad.usamc.mbx.pt@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This handbook provides detailed guidance to all personnel involved in assembly, handling, packaging, and storing of ESDS items, excluding ordnance and ordnance containing electronics. Items that contain ordnance assemblies may be considered an ESDS item; however, the static control and grounding required to ensure the safety of the item and ordnance operators are beyond the scope of this handbook. **This handbook is for guidance only and cannot be cited as a requirement**.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

FEDERAL SPECIFICATIONS

PPP-B-1672	-	Boxes, Shipping, Reusable with Cushioning
PPP-C-795	-	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film, for Long Distribution Cycles)
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

COMMERCIAL ITEM DESCRIPTIONS

A-A-3129	-	Cushioning Material, Flexible Open Cell Plastic Film (for Packaging Applications)
A-A-59135	-	Packaging Material, Sheet
A-A-59136	-	Cushioning Material, Packaging, Closed Cell Foam Plank

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117	-	Bags, Heat-Sealable
MIL-PRF-55182	-	Resistor, Fixed, Film, Non-established Reliability, Established Reliability, and Space Level, General Specification for
MIL-PRF-55342	-	Resistor, Chip, Fixed, Film, Nonestablished Reliability, Established Reliability, Space Level, General Specification for
MIL-PRF-81705	-	Barrier Materials, Flexible, Electrostatic Protective, Heat-Sealable

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MIL-DTL-81997 - Pouches, Cushioned, Flexible, Electrostatic-Protective, Transparent
MIL-PRF-83401 - Resistor Network, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed, Film Resistor, General Specification for
MIL-PRF-87893 - Workstation, Electrostatic Discharge (ESD) Control

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Standard Practice for Military Marking for Shipment and Storage
MIL-STD-2073-1 - Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
JEDEC SOLID STATE TECHNOLOGY ASSOCIATION
EOS/ESD ASSOCIATION, INC.

ANSI/ESD S20.20 - Development of an Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices)
JEDEC JESD625 - Requirements for Handling Electrostatic-Discharge-Sensitive (ESDS) Devices
ESD ADV1.0 - ESD Association Advisory for Electrostatic Discharge Terminology

(Copies of ANSI documents are available online at https://www.ansi.org or from the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036, telephone (212) 642-4900, facsimile (212) 302-1286 or Email: storemanager@ansi.org or info@ansi.org.)

(Copies of ESD documents are available online at https://www.esda.org or from the EOS/ESD Association, Inc., 7900 Turin Road, Building 3, Rome, NY 13440-2069, telephone (315) 339-6937, or Email: info@esda.org.)

(Copies of JEDEC documents are available online at https://www.jedec.org or from JEDEC, Global Standards for the Microelectronics Industry, 3103 North 10th Street, Suite 240-S, Arlington, VA 22201-2107.)

ASTM INTERNATIONAL

ASTM D996 - Standard Terminology of Packaging and Distribution Environments
ASTM/D1974 - Standard Practice for Methods of Closing,

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D1974M		Sealing, and Reinforcing Fiberboard Boxes
ASTM/D5118	-	Standard Practice for Fabrication of Fiberboard
D5118M		Shipping Boxes
ASTM/D5330	-	Standard Specification for Pressure-Sensitive Tape
D5330M		for Packaging, Filament-Reinforced
ASTM D5445	-	Standard Practice for Pictorial Markings for Handling of Goods
ASTM D/D5486	-	Standard Specification for Pressure-Sensitive Tape
D5486M	-	for Packaging, Box Closure, and Sealing

(Copies of ASTM International standards are available at https://www.astm.org or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959, telephone (877) 909-2786 (USA and Canada), (610) 832-9585 (International), or Email: service@astm.org.)

3. DEFINITIONS

"Definitions of all key terms used in the handbook should be stated in sufficient detail to provide for adequate clarity, unless the terminology is generally accepted and not subject to misinterpretation. Definitions may be included by reference to another document. When a standard definition exists, a different definition should not be used. Definitions are listed in alphabetical order. A list of acronyms and abbreviations may also be included under definitions."

3.1 Definitions. General definitions of terms unique to this standard are listed below. Definitions of other terms commonly used in the packaging community may be found in ASTM D 996 and ESD ADV1.0.

3.2 Antistatic. Refers to the property of a material that inhibits triboelectric charge generation effects.

3.3 Antistatic fast pack. Reusable container with antistatic cushioning and the word "antistatic" printed on both ends.

3.4 Antistatic tote tray. Conductive containers ideal for static-free storage, kitting, and in-process handling and transporting of ESDS assemblies or devices.

3.5 Conductive. Materials that have the ability to conduct electrical current. Such materials may be either metal or impregnated with metal, carbon particles, or other conductive ingredients or whose surface has been treated with such materials through a process of plating, metallizing.

3.6 Electrostatic discharge (ESD). The rapid, spontaneous transfer of electrostatic charge induced by a high electrostatic field. Note: Usually, the charge flows through a spark between two bodies at different electrostatic potentials as they approach one another. Details of such processes, such as the rate of the charge transfer, are described in specific electrostatic discharge models.

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3.7 ESD Protected Area (EPA). A defined location with the necessary materials, tools and equipment capable of controlling static electricity to a level that minimizes damage to ESD susceptible items.

3.8 ESD protective material. Packaging materials designed and tested to provide one or more of the following characteristics: minimizes triboelectric charge generation, dissipates electrostatic charges, shields electrostatic or electromagnetic field forces.

3.9 ESD protective packaging. An umbrella term to depict the use of ESD protective packaging materials as part of an overall ESD program to prevent damage due to ESD events.

3.10 ESDS items. Electronic items which are susceptible to damage or degradation as a result of an electrostatic discharge event.

3.11 ESDS labels. All unit packs, intermediate and exterior containers of ESDS devices (or other electronic items that are alone or are part of assemblies) that are susceptible to damage from ESD should be marked with an approved ESDS label (see page 21 for Approved Labels).

3.12 Intermediate pack. A wrap, box, or bundle that normally contains two or more unit packs of identical items.

3.13 Ionizer. A device that is designated to generate positive and/or negative ions. In an ESD protective workstation, it is designed to neutralize or eliminate an electrostatic field that could be hazardous to the handling or packing of ESDS items.

3.14 Static dissipative. For the purposes of ESD protective materials covered in this document: A material that has a surface resistance of greater than or equal to $1x10^4$ ohms but less than $1x10^{11}$ ohms or a volume resistivity of greater than or equal to $1x10^4$ ohm-cm but less than $1x10^{11}$ ohm-cm.

3.15 Static electricity. Electrical charge at rest. The electrical charge is due to the transfer of electrons within a body or from one body to another.

3.16 Triboelectric effect. The generation of static electricity caused by rubbing two substances together or by the contact and separation of two materials.

3.17 Unit pack. The first tie, wrap, or container applied to a single item or a quantity thereof or to a group of items of a single stock number, preserved, or unpreserved, which constitutes a complete or identifiable package.

4. GENERAL INFORMATION

4.1 Generation of electrostatic charges. Electrostatic charges are generated by the contact and separation of the surfaces of two materials, with at least one being nonconductive. Friction caused by rubbing materials together tends to increase the amount of accumulated electrostatic charges. A familiar example of ESD is the spark that occurs by walking across a carpet and touching a metal doorknob. The contact and separation of shoes and carpet generate an electrostatic charge that is induced on a person's body, which is sufficiently conductive due to

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salty moisture on the body's surface. Since the doorknob is metal and highly conductive, a spark occurs when the hand is brought close to the doorknob, assuming the electrostatic charges are great enough.

4.1.1 Dry and moist materials. Separating dry materials generates greater electrostatic charges than moist materials because the moistness is sufficiently conductive and helps to dissipate the charge. For this reason, ESD effects are more noticeable in the winter since heating systems reduce moisture on the surfaces of furniture and other objects. Any circumstance that results in a low relative humidity will permit a greater accumulation of electrostatic charges.

4.1.2 Static charge. A static charge is an excess or a deficiency of electrons on a surface. An excess of electrons produces a negatively charged surface and a deficiency of electrons produces a positively charged surface. Figure 1 gives examples of common materials that charge positively, or negatively when separated from each other. Positively charged materials lose their loosely bound surface electrons and negatively charged materials attract and maintain those electrons.

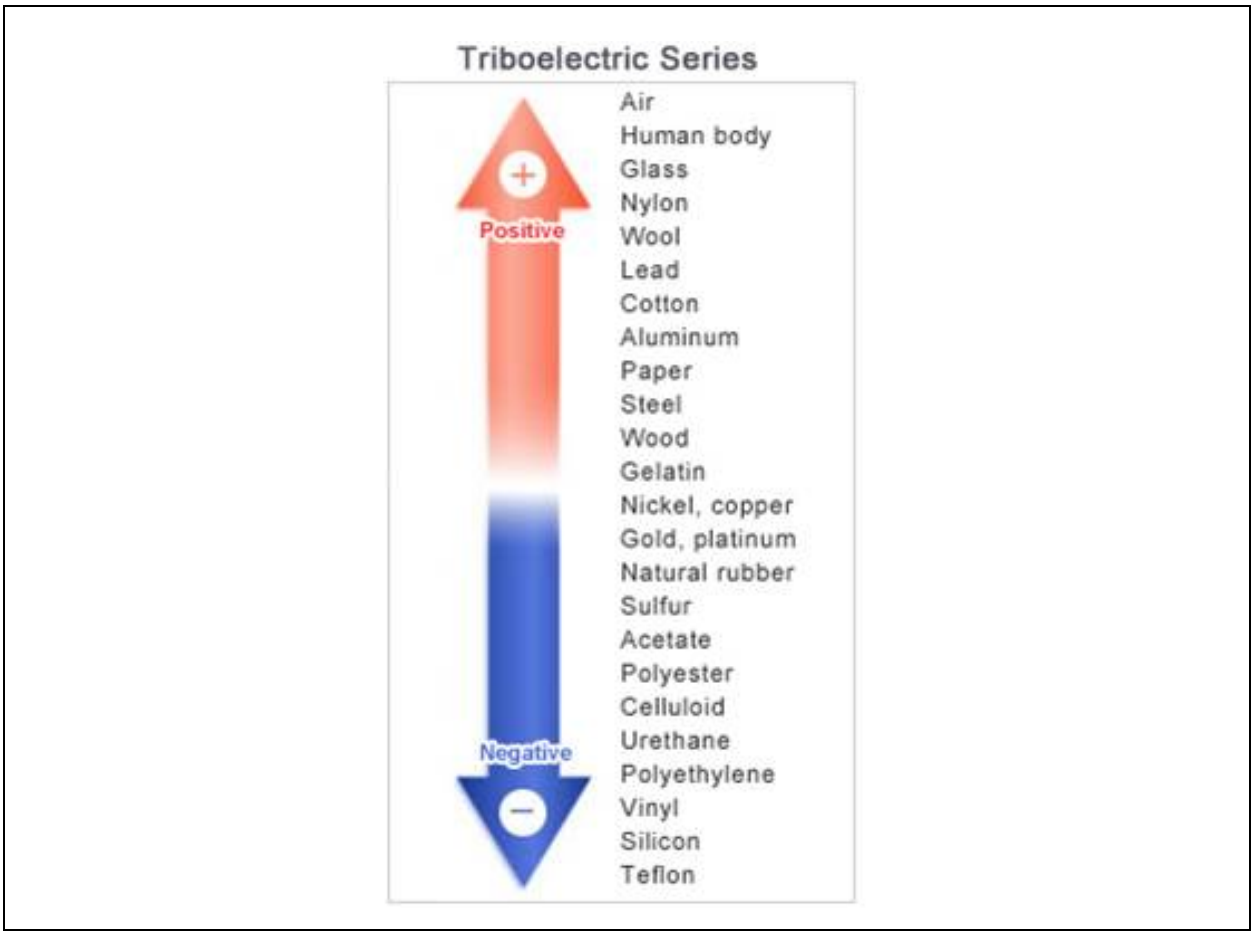


FIGURE 1. Common materials that generate triboelectric effect.

Table I shows typical electrostatic voltages that occur during normal workday activities.

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TABLE I. Typical electrostatic voltages.

Means of static generation	Measured voltages	
	10% to 20% RH	65% to 90% RH
Walking across carpet	35,000	1,500
Walking over vinyl floor	12,000	250
Worker at bench	6,000	100
Vinyl envelopes for work instructions	7,000	600
Common poly bag picked up from bench	20,000	1,200
Work chair padded with polyurethane foam	18,000	1,500

Table II lists typical electrostatic charge sources.

TABLE II. Sensitivity ranges.

Component	Volts
Transistors	30 - 7,000
Operational amplifiers (modules)	200 - 2,500
Diodes	300 - 2,500
Film resistors	300 - 3,000
Low-power, integrated circuit logic	500 - 2,500
Silicon controller rectifier	700 - 1,000

4.2 ESD damage. As shown in table II, activities such as walking, working at a bench, and handling common materials can generate thousands of volts of static electricity. If a spark discharge occurs from an accumulation of electrostatic voltage, electronic components, printed circuit board components, and other electronic assemblies can be destroyed or damaged (see figure 2). Spark discharges cannot be felt if the electrostatic voltage is less than the threshold level for that particular individual. This threshold level is usually within the range of 2,500 to 4,000 volts of electrostatic charge.

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Failure site indicates damage to surface mount capacitor.



Closer view indicates “punch-through” due to ESD.

FIGURE 2. Magnified views of ESD caused failures.

4.3 Description of an integrated circuit. Since the advent of the integrated circuit, electronic components have become smaller and their functional capabilities have become greater. A microprocessor, the core element in a microcomputer, is less than one-half of a cubic inch in volume. Because of the micro-miniature size of the functional components in an integrated circuit, they are much more susceptible to ESD damage than their earlier, significantly larger counterparts. Table III provides the relative ranges for possible damage.

TABLE III. Typical electrostatic charge sources.

Object or process	Common materials or activities
Work surfaces:	Waxed, painted, or varnished surfaces
	Vinyl or plastics
Floors:	Sealed concrete
	Waxed, finished wood
	Rugs
	Vinyl tile or sheeting
Clothes:	Clean room apparel
	Animal hair-based garments
	Nonconductive shoes
Chairs:	Finished wood
	Foam cushions

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Object or process	Common materials or activities
	Vinyl, fiberglass
	Plastic casters
Packaging and handling:	Plastics (bags, wraps, envelopes, labels, stretch and shrink films, tapes
	Bubble pack, foam, loose fills plastic trays, pallets, tote boxes, vials, parts, bins, containers)
Assembly, cleaning, test, and repair areas:	Spray cleaners
	Plastic solder suckers
	Solder irons with ungrounded tips
	Cleaning or drying by fluid or evaporation
	Heat guns and blowers
	Sandblasters
	Electrostatic copiers

4.3.1 Damage by spark discharge. Considering the threshold levels at which most people can feel a spark discharge from electrostatic voltage, it is easy to understand how electronic parts can unknowingly be damaged during handling. If it is not immediately destroyed, the damaged part may not cause a functional system failure until after it reaches the user.

4.4 Identification/acquisition. Items identified as ESDS should be protected from the very beginning of the acquisition process to final disposal. Specific control programs involved with item design, test, inspection, manufacture and maintenance are contained in ANSI/ESD S20.20.

4.4.1 Special protective devices. The requirements for the use of special protective devices prior to packaging, as with shunting bars, clips, or non-corrosive conductive foams for protection, is the responsibility of the Government activity involved in item acquisition.

5. DETAILED GUIDANCE

5.1 Identifying ESDS items. Due to the tremendous volume of electronic parts presently used by the military and the methods needed to accomplish the task, identifying parts that are ESDS is essential. ESDS items are categorized as follows:

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a. A FLIS/FEDLOG query for National Stock Numbers (NSNs) contains a data field identified as “ESDC” (Electrostatic Discharge Code). This is a one character alpha code (for example, A, B, C, or D) that indicates whether an item is susceptible to ESD or electromagnetic interference damage. If there is no code in this field, do not assume that item is not ESDS. The owners of ESDS items may have not entered a code in this data field.

b. If the letters “GX” are displayed in the MOP (Method of Preservation) field in an NSNs FLIS/FEDLOG catalog data, it is an ESDS item (see MIL-STD-2073-1, Appendix J, table J.Ia, Specialized preservation codes).

c. All printed circuit cards and wiring boards with mounted ESDS components; i.e., commonly included in Federal Supply Class (FSC) 5998.

d. All parts within FSC 5962.

e. The U.S. Army Aviation and Missile Command has identified FSCs 5955, Oscillators and Piezoelectric Crystals, 5963, Electronic Modules, and 7042, Mini and Microcomputer Control Devices as ESDS items.

f. The DLA Land and Maritime has identified specific resistors and semiconductors listed below as ESDS items.

(1) FSC 5905 resistors:

(a) Film resistors conforming to MIL-PRF-55182.

(b) Resistor chips conforming to MIL-PRF-55342.

(c) Resistor networks conforming to MIL-PRF-83401.

(2) FSC 5961 semiconductor devices:

(a) Microwave diodes.

(b) Insulated gate field effect transistors (Metal Oxide Semiconductor Field Effect Transistors (MOSFET).

(c) Junction field effect transistors.

(d) Silicon-controlled rectifiers.

(e) Hybrid semiconductors.

(f) Small signal Schottky diodes.

(g) Semiconductor devices, not otherwise named, which operate at a frequency above one gigahertz.

g. A number 3, used to specify a Type Cargo Code (TCC), can be found in the FLIS/FEDLOG data. TCC is a one character alpha/numeric code used for all shipments via water to

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identify the type of cargo included within the shipping unit. The number 3 identifies ESDS items, for example, material subject to damage caused by ESD.

5.2 Packaging of ESDS items. In all cases, the first contacting layer (with bare item) of the selected protective material will be static dissipative and not conductive. Such a layer may be in the form of a homogeneous single film or cushion thickness, or a multi-layered structure either as a flat film or cushioning pad as in a pouch configuration. Shielding will be required to protect against the effects of static discharge. This protection is normally available in materials through the use of conductive layers attained by carbon loading, metallization or other manufacturing techniques as provided by MIL-PRF-81705. Selection of the appropriate preservation materials will be based on the nature of the item and prescribed packaging data. MIL-STD-2073-1 identifies procedures for the packaging of ESDS items with the specialized preservation code GX in table J.Ia, and marking requirements via a special marking code, 39, found in table J.X. Unpreserved ESDS items should be handled within EPAs by trained personnel who can effectively employ the protective materials and equipment provided. Only approved ESD protective materials will be used to form the unit pack. Examples of code GX preservation are shown in figure 3 and figure 4.



MIL-DTL-81997D, type I



Bag marked IAW MIL-STD-129R



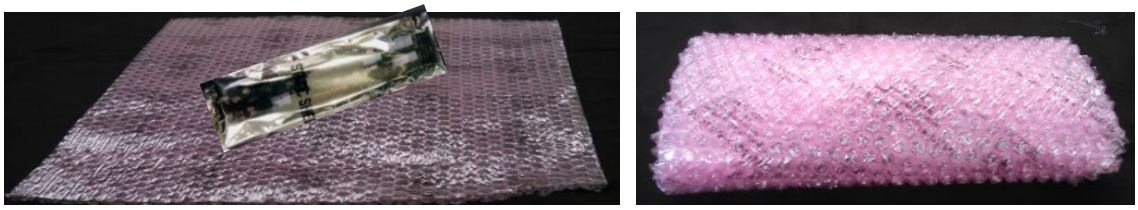
MIL-DTL-117H, type I, class F, style 1 bag (MIL-PRF-81705E, type I material)

FIGURE 3. GX using MIL-DTL-81997D, type I pouch.

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MIL-PRF-81705E, type III or
MIL-DTL-117H, type II, class H, style 2



A-A-3129C, type I, grade B



MIL-DTL-117H, type I , class F, style 1 Bag
(MIL-PRF-81705E, type I material)

FIGURE 4. GX using A-A-3129C, type 1, grade B.

a. Either wrap the item in material conforming to MIL-PRF-81705 type III; or place in a MIL-DTL-117, type II, class H, style 2 bag. Once wrapped, cushion in material conforming to A-A-3129, type I, grade B or PPP-C-795, class 2 or MIL-DTL-81997, type II or PPP-C-1797, type II or A-A-59136, class 1, grade B or MIL-PRF-26514, type III, class 2, to protect item and to prevent bag puncture. Alternately if applicable, re-closable cushioned pouches conforming to MIL-DTL-81997, type I may be used in lieu of initial wrap and cushioning.

b. Place the wrapped and cushioned item in a barrier bag meeting the criteria specified in MIL-DTL-117, type I, class F, style 1, or bag constructed from MIL-PRF-81705, type 1, class 1 or 2 material. The bag should be made of one piece of material folded in half and heat sealed on three sides. This becomes the unit pack holding the ESDS item and can now be handled as you would any non-ESDS item.

c. Place an ESDS device unit pack label (see figure 4) on the unit pack. In the absence of NSNs, local purchase procedures should be followed when ordering this unit pack ESDS label. Should this label be temporarily unavailable, unit packs may be marked on two opposing

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sides with the ESDS device symbol and the statement: "DO NOT OPEN – EXCEPT AT APPROVED ESD PROTECIVE AREA".



ESDS label
(Unit Pack)

NSN	OPTIONAL FORMS
7540-01-317-7371	87



ESDS label
(Intermediate and Exterior Packs)

NSN	OPTIONAL FORMS
7540-01-109-8815	87

FIGURE 5. ESDS Labels.

d. Once properly preserved as above, packing for ESDS items can be accomplished using ASTM D5118/D5118M fiberboard containers. Allow for ESD protective cushioning (as necessary). Refer to the recommended fiberboard most frequently used container list specified in MIL-STD-2073-1. The listed sizes may be requisitioned through normal supply channels. If box-making equipment is available, containers should be fabricated in accordance with ASTM D 5118/D 5118M. Closure should be in accordance with ASTM D1974/D1974M. Lastly, mark and apply ESDS labels (see figure 4) in accordance with MIL-STD-129.

e. Alternate packing for repairable ESDS, reusable PPP-B-1672, type II, style D, folding convoluted "fast packs," may be specified for use. There are size limitations for PPP-B-1672 and service prescribed packaging instructions should be consulted prior to use.

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5.3 Transporting before final unit pack. If the unit pack needs to be opened for any purpose, it should be opened at an EPA (see figure 6). If the item needs to be transported to several areas before it is converted to a unit pack (for example, from receiving inspection, to repair and test, to inspection, then to final unit pack), use the following procedures:

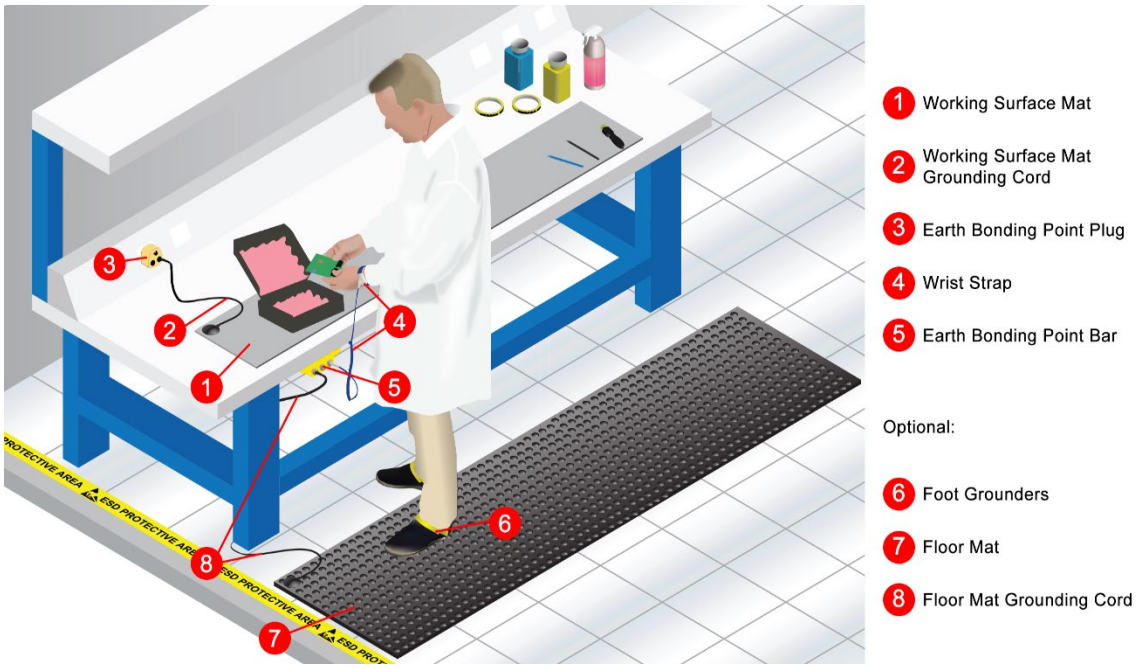


FIGURE 6. Example of ESD protective workstation.

a. Wrap, cushion, or pouch the item in accordance with 5.2.

b. Place the wrapped or cushioned item in an open bag conforming to MIL-DTL-117, type I, class F, style 1. MIL-PRF-81705, type I, class 1 or 2. Note: some services may allow maintenance organizations to transport items utilizing MIL-DTL-117, type II, class H, style 2 bags during in-house maintenance prior to packaging final unit pack.

c. Fold the open edge over and use antistatic adhesive tape or ESDS label to hold the flap down. Common adhesive tape should not be used, since it can generate an electrostatic charge when removed from the roll, or the bag.

5.4 Condition code F material. All reparable ESDS items should be rendered the level of packaging that would prevent further deterioration of the item. Returned condition code F material should be given the same applicable packaging referenced on the preceding pages beginning with 5.2, for example, condition code A, repaired, or newly procured ESDS items.

5.5 ESD protective packaging materials. There is a Qualified Product List (QPL) for MIL-PRF-81705 ESD barrier materials (QPL-81705), that is located in the Qualified Products Database (QPD). Go to https://quicksearch.dla.mil/. On this page are the words: Document Number. Type in the numbers “81705”. Click on “Search”. This action brings up another page, look for and click on “QPL-81705”. Click on “View QPD data”. This action brings you to a QPL document listing of ESD packaging materials and approved suppliers. To view the current

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MIL-PRF-81705 qualified product list of approved manufactures and materials, click on the hyperlinks for each material type listed.

5.6 EPA. Handling of ESDS items, parts, assemblies and equipment without ESD protective covering or packaging should be performed while in an EPA. The EPA should have clearly identified boundaries. An EPA can consist of a single workstation, entire room, building or other designated area. Access to the EPA should be limited to personnel who have completed appropriate ESD training. Untrained individuals should be escorted by trained personnel while in an EPA. An EPA should be established wherever ESDS items are handled. However, there are many different ways to establish ESD controls within an EPA. Figure 6 is an example of an ESD protective workstation, which depicts typical items that are important in handling ESDS parts safely. The ionizer, when required, is a precautionary device that, within its effective area of control, will aid in removing electrostatic discharges from isolated conductors and nonconductive surfaces. Depending upon service requirements, heel straps and a static dissipative floor mat may also be included with the essential worksite items. Heat-sealing equipment may be located adjacent to the workbench so that the wrist strap remains attached while heat-sealing bags. The tabletop and floor mat ground wires should each go to ground directly (see figure 6, items 3 and 5). To prevent the possibility of shocks from high-voltage sources, wrist straps for use at ESD protective workstations contain a protective resistor (usually one megohm). Grounding points should be checked when installed, relocated, or as required per service guidance.

5.7 Precautions. Observe two basic rules whenever possible: Handle all ESDS items at an EPA and transport all ESDS items in static protective containers or packages. Adhere to the following precautions when using an EPA.

a. When protective apparel is worn it should be frequently checked, especially after cleaning, by scanning personnel with an electrostatic field meter to monitor for damaging ESD voltages.

b. Be sure to attach the wrist strap (mandatory) and heel straps (if used) before handling any ESDS items.

c. Avoid the presence of any non-antistatic or insulative material on or near the worksite and workstations such as Styrofoam cups, common plastic or masking tape, common wrapping or barrier materials (for example, bubble pack, or plain poly) cigarette packs, and synthetic materials (for example, rayon, orlon, or plastics of any type) or any other non-job insulating materials.

d. Do not store or use magnetic material near the worksite.

e. Do not perform stretch- or shrink-wrapping operations within the ESD-controlled work area.

f. If more than one EPA is located in the same area, do not connect the tabletops or floor mats grounding points in series; individually ground each one.

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g. Do not use waxes, polishes, or similar material on the floor mat or tabletop. They may deposit an insulating layer of residue reducing or eliminating the effectiveness of the floor mat or tabletop. For the same reason, the use of topical antistatic spray is not recommended on packaging materials.

h. Sweep dust and dirt from the tabletop and floor mat as often as needed to prevent any accumulation which will also insulate them and make them ineffective.

i. Perform continuity checks of all ground wires weekly (or as directed by manufacturer or Service requirements) to ensure that the proper grounding is maintained. A log of grounding point system checks should be maintained for documenting when checks were performed. The log should be available for reviews by supervisors and Quality Assurance personnel. (See ANSI/ESD S20.20 and JEDEC JESD625).

j. Check wrist straps and foot grounding devices daily to ensure that the protective resistor (usually one megohm) is still in the circuit and that personnel are properly grounded. This is typically accomplished during a wrist strap system test (wristband, ground cord, and person). If the resistance is less than 750K ohm (or service defined value), replace defective component. If the resistance is greater than 10M ohm (or service defined value), correct fit or replace faulty component.

k. ESD protective pouches should be used for technical manuals, drawings, work instructions, etc., rather than plain plastic pouches, if they are used at all, or keep these items at least 1 feet from ESDS items when possible.

5.8 Typical ESD protective workstation. Figure 6 shows a typical ESD protective workstation (the heat-sealing equipment is not shown). Indicated on the figure are key ESD-control components that comprise a workstation. Figure 7 depicts several close-up examples of these components.

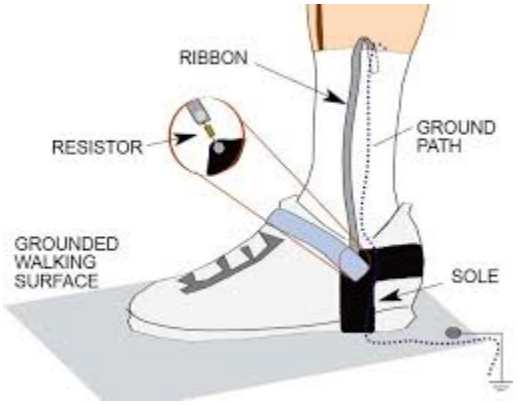
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WRIST STRAP WITH RESISTOR





FOOT GROUNDER

FIGURE 7. EPA personnel grounding (Service requirements may vary).

5.9 ESD protective workstation and EPA materials. The work performed in the EPA will typically determine the type of work surface material best suited for the application. If the items being handled are sensitive to mechanical shock, then a cushioning material may be required. If heavy chassis are moved around the work surface, then a durable surface may be necessary. High voltage operations may require special considerations for the electrical properties of the work surface to ensure personnel safety. Work surfaces used in cleanrooms may also require special consideration, including particulate contamination and outgassing properties. Actual resistance values chosen within the EPA for work surfaces, work surface ground cords, wrist straps, chairs, floor, or other ESD control items, when required, are typically chosen to ensure static charges are safely dissipated to ground.

5.10 ESD field service kit. ESD protection is required where formal workstations are not practical (that is, remote locations such as shelters and vans). ESD control field service kits primary use is when removing ESDS items from service or place them in operation during which time the items are not inside a standard EPA. Figure 8 reflects a typical kit as described in Appendix A. It is noted that these static dissipative mats with accessories are not intended as a substitute for a permanent ESD protective workstation. The kit is field expedient (portable) and meets minimum requirements for an ESD protective workstation for these remote applications.

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FIGURE 8. Typical ESD protective field service kit.

6. NOTES

6.1 Intended use. This handbook provides guidance on procedures, materials, and practices that can be used to protect and extend the service life of ESDS items. Information is provided on how to receive, handle, pack, mark, and ship ESDS items.

6.2 Subject term (key word) listing.

ESD protected area
ESD protective material
ESD protective workstation

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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ELECTROSTATIC DISCHARGE (ESD) FIELD SERVICE KIT

A.1 SCOPE

A.1.1 Scope. This appendix provides, basic instructions on how the ESD field service kit (MIL-PRF-87893, type III) should be used to handle and protect ESDS items.

A.2 PROCEDURE

A.2.1 ESD protective field service kit instructions. The following steps should be followed when working with an ESD protective field service kit.

CAUTION: Only trained personnel should handle unprotected ESDS items. Disconnect all power before handling any item.

STEP 1. Place unfolded work mat on table or floor near work area or service location.

NOTE: Remove all ESD packaging material from pockets and place away from mat.

STEP 2. Attach ground cord to mat fastener. Attach clip to mainframe, chassis, or other bare metal surface near item to be replaced or removed. This surface has to be a part of unit to be replaced or removed, and be a part of unit assembly containing item.

STEP 3. The wrist strap coiled ground cord will be attached to the built-in ground point on the mat's ground cord. Adjust wrist strap over wrist against bare skin.

STEP 4. Proceed now to remove, replace, or handle the ESDS item only on the mat surface. Keep all other non-job essential items off the mat.

STEP 5. ESDS items to be returned (for example, condition code F) for repair or otherwise removed from the static dissipating mat should be packaged to prevent any further deterioration (see paragraphs 5.2 and 5.3).

NOTES:

1. Do not disassemble the field service kit until all procedures have been completed.
2. Ground cord and wrist strap cord are usually connected together on the kit's ground snap(s).
3. Follow manufacturers test method and frequency (or Service specific requirements) for evaluating the electrical and mechanical characteristics of wrist strap and ground cord.

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CONCLUDING MATERIAL

Custodians:
Army – SM
Navy – AS
Air Force – 20
DLA – DH

Preparing Activity:
Army – SM
(Project PACK-2020-008)

Review Activities:
Army – AR, AT, CR, MI
Navy – MC, OS, SA, SH, YD, CG
Air Force – 70, 71, 170, 69
Other - MDA

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